Frontier Technical Reconnaissance — Circular Materials, Children's Apparel Textile Circularity, and Advanced Plastics Recycling (June 2026)

TL;DR

- The single highest-shock-per-effort intersection in textile circularity right now is **elastane** — the 2–5% polyurethane elastomer that contaminates virtually all stretchy infant/children's apparel, defeats every commercial recycling stream (mechanical, glycolysis, enzymatic, pyrolysis), and has *no* commercial-scale selective-removal solution; the most advanced lab work (DMSO/DBN selective degradation, chitosan dissolvable finishes, Samsara Eco's AI-engineered enzymes, Carbios's enzymatic depolymerization with "hard-point" preparation lines) is still pilot-scale and the published yield/selectivity numbers leave room for genuine breakthroughs.
- The Cox/Nexus pyrolysis thesis sits on top of three openly contested technical seams — (i) the ~15–30% real plastic-to-plastic yield once you account for cracker losses (vs. the 80–85% pyrolysis-oil yields companies report), (ii) the chlorine/PVC/heteroatom contamination ceiling that prevents film-derived oil from becoming food-grade or fiber-grade polymer at scale, and (iii) the "free allocation / fuel-use exempt" ISCC mass-balance accounting controversy now in active litigation (California AG v. ExxonMobil). The frontier where software/AI most plausibly substitutes for chemistry is **catalyst discovery** (Sadow/Ferrandon's 2026 *ACS Catalysis* ML-discovered bimetallic hydrogenolysis catalysts) and **inverse-design of dechlorination/depolymerization catalysts**.
- For a hackathon team chasing maximum technical ceiling, the densest unsolved seams are: (1) elastane-tolerant fiber-blend deconvolution at the molecular level; (2) source-level microfiber-shedding mitigation in soft contact-with-infant fabrics (no benchmark, no standard, the As You Sow Target/Gap 2026 shareholder resolutions explicitly identify this as a "viable solutions exist but no measurement framework" zone); (3) physical-layer traceability that survives 100+ wash cycles AND chemical recycling (Haelixa's DNA-marker survives spinning/dyeing/finishing but is melt-destroyed); (4) AI-driven inverse design of enzymes/catalysts for the specific contaminant cocktail in children's apparel (dye + finish + elastane + PFAS legacy); (5) robotic soft-object manipulation for in-line hard-point removal at MRF speeds. Honest assessment of attack vectors is in each section below.

Key Findings

1. The frontier is bottlenecked by elastane, not by enzymes. Carbios, Samsara Eco, and Circ have working enzyme/chemistry platforms for PET and (in Samsara's case) nylon 6,6, with commercial plants opening 2025–2027. None has a published commercial solution for the 2–5% elastane that contaminates infant/children's stretch apparel. The Chavez-Linares 2025 review in *Global Challenges* and the Brussels/Ghent DMSO-DBN work (which achieved 81–100% elastane mass loss at 80–120 °C with 0.1% v/v DBN catalyst) are the published frontier — still pilot/lab.

2. The 2025 polycotton-to-glucose breakthrough is real and explicit about its remaining gap. Leenders/Gruter et al. (*Nat. Commun.* 16, 738, 2025) achieved 75% molar glucose yield from cotton in 44/56 polycotton at room temperature with 43 wt% HCl, scaled at 1 mL, 0.1 L, 1.0 L and a 230 L Avantium pilot reactor. The polyester residue is then glycolyzed at 1:16 PET:EG with 1% Zn-acetate, 11 h reflux. The unsolved seam stated by the authors themselves: this assumes a clean polycotton; **elastane, dyes, finishes, and trims are not handled** in the published process and would need an upstream selective-removal or DNA/molecular sorting step. Nature Nature

3. The advanced-recycling (pyrolysis) thesis Cox is backing has three openly contested technical seams. Nexus Circular's published yield is "just over 80%" pyrolysis oil from HDPE/LDPE/PP/PS — but this is mass to oil, not mass to new plastic. Per Lisa Song, "The Delusion of Advanced Plastic Recycling Using Pyrolysis," *ProPublica* (Jun 2024): "if a pyrolysis operator started with 100 pounds of plastic waste, it can expect to end up with 15–20 pounds of reusable plastic," with 25–30 pounds for more advanced technologies, because the downstream steam cracker only converts <50% of naphtha-equivalent oil into the olefins that become new polymer. Zero Waste Europe's *Leaky Loop Recycling* (Oct 2023) drew on 22 peer-reviewed papers and concluded the end-to-end recycled-plastic yield "is at most 2%" — contested by Chemical Recycling Europe but never quantitatively rebutted. Brightmark's Ashley, IN plant (100,000 t/y nameplate) operated at five percent capacity per its March 2025 Chapter 11 filing — a real-world demonstration that scaling pyrolysis is genuinely hard. ACS C&EN + 4

4. The chlorine/PVC heteroatom ceiling is the real technical seam in film pyrolysis. Lee et al. (*Phil. Trans. R. Soc. A* 2025): "A safe and promising PVC dechlorination method is urgently needed... none has been proven at the commercial scale." HCl from PVC is autocatalytic and poisons downstream zeolite catalysts (Pt/ZSM-5 reported destroyed in *Sci. China Chem.* 2024). DKR-350 batch pyrolysis study (Strien et al., *Energy & Fuels* 2024, 10.1021/acs.energyfuels.4c04745) achieves 78% oil yield with dry-washed feed but the oil contains chlorine, nitrogen (from PA/PU), and sulfur. The 2024 *RSC Chem. Sci.* (10.1039/D4SC00130C) silylium-ion full-PVC-dechlorination paper converted PVC to branched polyethylene in 2 h — lab only. nih + 3

5. Catalytic upcycling of polyolefins is the closest thing to a software-can-attack-a-chemistry-problem inflection point. Susannah Scott (UCSB), Aaron Sadow (Ames Lab, iCOUP DOE Energy Frontier Research Center), and John Hartwig (UC Berkeley) have demonstrated solid-catalyst routes from PE/PP directly to high-value monomers/aromatics at 90% lab yields. Conk/Hartwig *Science* 2024 (DOI: 10.1126/science.adq7316): Na/Al_2O_3 + WO_3/SiO_2 at ~90% yield to propylene + isobutylene. Sadow/Delferro's $Pt/SrTiO_3$ (*ACS Cent. Sci.* 2019) at 300 °C, 170 psi H_2. The **2026 ML breakthrough** (Byron/Sadow/Delferro/Ferrandon, *ACS Catalysis* 16(1):431-

445, 2026): data-driven discovery of bimetallic nanoparticles for PE hydrogenolysis — the first explicit ML-discovered plastic-deconstruction catalyst. This is the highest-ceiling intersection: AI-driven inverse catalyst design. ACS C&EN + 4

6. NIR/hyperspectral sorting has a published, hard accuracy ceiling specifically on the cases that matter most for kids' apparel. Refiberd publicly claims detection of "spandex, nylon, and acrylic below 2% composition" — but multiple published papers (*ScienceDirect* S1386142525009722, 2025; MDPI *J. Imaging* 2025, 11(4):96) confirm that NIR struggles fundamentally with dark/black textiles (because carbon black absorbs across the NIR window) and with elastane at the trace levels typical of children's wear (1–5%). The cost/performance gap is real: hyperspectral systems cost ~5× NIR cameras and remain at the industrial-only price point, while handheld Matoha devices accept the accuracy loss. There is no published benchmark dataset for blend-quantification on contaminated/colored/worn infant textiles.

7. Robotic soft-object manipulation has had two distinct 2024–2025 breakthroughs, but garment disassembly remains unsolved. APS-Net (ICML 2025; arXiv 2506.22769) demonstrated dual-arm garment unfolding + standardization in simulation/limited-real. Multiple soft-robotic dressing systems are at TRL 3–4. **No published system performs automated removal of zippers, snaps, buttons from real, worn children's garments at industrial throughput.** Carbios's textile preparation line at Clermont-Ferrand integrates shredding + hard-point extraction in an automated industrial line — but this is a chopping-and-mechanical-removal system, not a soft-manipulation deconstruction system.

8. Microfiber shedding has shifted from filter-side to source-side, but there is still no standardized measurement. The 2025 Taylor & Francis review (Khan et al.) confirms: "The absence of universally adopted protocols for measuring microfiber release during laundering and other life cycle stages has resulted in data inconsistency and incomparability across studies." Toronto's Golovin lab is working on a single coating that adheres to multi-fiber substrates (polymer-architecture challenge explicitly noted). Polysilk-CTE silicone finish has shown 63.25% reduction on polyamide. France's mandatory washing-machine microfiber-filter law took effect Jan 2025; the As You Sow shareholder resolutions on Target (2026 filing, "26TGTCE") and Gap (Dec 2025) explicitly identify the lack of harmonized testing as the blocking issue for setting brand reduction targets. Under Armour committed 75% low-shed by 2030. The infant-skin angle is real: dermal contact + thinner-skin permeability + 12–16 hours/day exposure puts kids' apparel at the front of this exposure pathway. ResearchGate

9. PFAS contamination of recycled textile feedstock is regulatorily acute for kids' products. The 2022 Whitehead et al. study (*Environ. Sci. Technol. Lett.*) found PFAS in 100% of stain-resistant children's textiles tested. The Whelan et al. 2024 EU rapid-extraction method now allows UPLC-TripleTOF/MS quantification of 21 PFAS in polymer waste streams at 0.1–0.4 mg/kg limits of quantification. **CPSIA does not currently mandate PFAS testing**, but California, NY, Maine, Washington, Oregon (TFKA) are layering PFAS-on-textile restrictions for 2024–2027 effective dates, and the EU's POP regulation imposes low-POP concentration limits on recycled material containing legacy PFAS — meaning recycled feedstock from pre-2024 children's clothing carries forward as a regulatory landmine. PubMed Central

10. Physical-layer traceability that survives wash + chemistry is uniquely owned by Haelixa. Haelixa's encapsulated DNA marker (ETH Zürich spinout, 2016) is verified to survive spinning, weaving, dyeing, bleaching, finishing, and multiple wash cycles when PCR-detected, and is now in production with New Focus Textiles (Nov 2025), Soorty (Pakistan, 2024), C&A, Hugo Boss, OVS. Haelixa explicitly states: "Gold melting and surface layer removal are the only industrial treatments that the markers do not survive." **The frontier gap: DNA markers will not survive enzymatic depolymerization or pyrolysis. So a DNA-marker that survives the wash cycle but is destroyed in chemical recycling is a partial solution.** No commercial tag survives both 100+ washes *and* chemical recycling. This is a genuine open seam. Resource

11. Carbon-negative / CO_2-derived fibers are real but at low TRL. LanzaTech's CarbonSmart™ polyester (from industrial-emissions-derived ethanol via FENC) is in commercial Lululemon, On, Zara, H&M Move, REI, adidas products as of 2025. Twelve + LanzaTech announced CO_2-to-polypropylene in 2024 (still pre-commercial). Newlight (AirCarbon, methane-fermentation PHA biopolymer) is in apparel pilots. Spiber's brewed-protein fibers (precision fermentation) launched with North Face and Pangaia at small commercial volumes. Keel Labs' Kelsun (kelp-derived alginate fiber) is at TRL 6–7. Colorifix's microbial dyes — $18M Series B in June 2025 led by Inter IKEA Group, with H&M Group Ventures as continuing investor (per *Just-Style*, Jun 24, 2025: "Colorifix secures $18m to rollout sustainable dyeing process [in a round] led by Inter IKEA Group") — hold the OEKO-TEX Eco Passport with biocompatibility tests, first biological-dye safety pioneer at TRL 7. **The CPSIA/REACH frontier gap is that next-gen materials have not been systematically lead/phthalate/AZO-tested for infant-skin-contact specifically — there's no published toxicology framework for "novel biomaterials in infant apparel."** Lanzatech + 3

12. The rebound/Jevons effect is the systems-level open problem that no AI demand-forecasting system has touched. Yerushalmi et al. 2025 (*Business Strategy and the Environment*, 10.1002/bse.70135) ran a multi-region multi-sector DCGE model on circular textile innovations and found a **155% rebound backfire** — circular-economy innovation in textiles/clothing, modeled honestly, makes environmental pressure worse, not better, because efficiency lowers cost and triggers latent demand. This is the open systems-level frontier where software *can* attack the problem at scale.

Details

Frontier 1 — Selective Fiber-Blend Separation & Recycling

State of the art (2025-2026):

- **Carbios** (Clermont-Ferrand, FR): PET-specific engineered hydrolase. Industrial demonstrator running since Sept 2021 (2 t/batch reactor). Longlaville commercial plant (50,000 t/y) construction resuming H2 2025 subject to funding (€72M cash on hand mid-2025). Operates after a patented automated textile preparation line that removes zippers/buttons. Targets PET-rich textile feedstock; cotton residue is left intact. **Limitation:** elastane/spandex is not addressed. Carbios + 2
- **Samsara Eco** (Jerrabomberra, AU): Opened first commercial nylon-6,6 + polyester enzymatic plant 2025. AI-engineered enzymes for PET, nylon 6,6, nylon 6 at low temperatures. EosEco brand. Handles "poly/cotton and nylon/elastane blends." 10-year Lululemon offtake for ~20% of fibre portfolio. **Their dye/finish/elastane challenge is the explicit subject of the Deakin REACH research partnership (Aug 2025).** Packagingconnections Knittingindustry
- **Circ** (Danville, VA): Hydrothermal process recovers both polyester and cotton from polycotton blends; partner with Selenis and Inditex; 70,000 t/y French flagship (announced via Macron's Choose France Summit 2025) operational target 2028. Patagonia, Zalando, Inditex investors. Trellis + 3
- **Ambercycle** (Los Angeles): Cycora regenerated polyester; MAS Holdings, Hyosung TNC, Reformation, Arc'teryx, Gap/Athleta partnerships. Trellis
- **Syre** (Stockholm, H&M-backed): 10,000 t/y "blueprint" plant Cedar Creek, NC, opening 2025; gigascale (150,000–250,000 t/y) target 2027. Series A $100M (2024). Volvo, TPG Rise, H&M ($600M offtake). Sourcing Journal + 3
- **DePoly** (Sion, CH): Room-temperature, ambient-pressure PET → TPA + MEG. 50 t/y pilot since 2020; 500 t/y showcase plant in construction. Tolerates colored/mixed/contaminated PET without pre-cleaning. $13.8M seed co-led by BASF VC + Wingman Ventures. Depoly + 2
- **RE&UP** (Türkiye / Sanko Textile): Targeting 200,000 t/y by 2025; 1 Mt/y by 2030. Polycotton focus. H&M Foundation
- **The 2025 Nature Communications work** (Leenders/Gruter/Avantium): 43 wt% HCl, room temp, 24 h → 75% molar glucose yield from cotton + intact PET residue. Scaled to 230 L pilot. The polyester is then recycled by glycolysis (1:16 PET:EG, 1% Zn-acetate, 11 h reflux) → BHET. **First effective process for full recycling of both components.** Nature Avantium

Unsolved seam: Elastane (next section). Multi-fiber cocktails (e.g., cotton/polyester/nylon/elastane with PU coating, common in baby outerwear). Dye/finish interference with enzyme kinetics. **Selectivity at the polymer level still requires upstream sorting at the garment level**, which is itself unsolved (Frontier 5).

Attack vector honest assessment: This is fundamentally a *wet chemistry / enzyme engineering* frontier. AI's role is in (a) Samsara/Carbios-style directed evolution of enzymes (AlphaFold-class structural prediction + in silico mutagenesis for protein engineering — this is the Samsara "proprietary AI platform"), and (b) catalyst inverse design (next section). A pure software attack here looks like a wet-lab co-pilot: in silico-evolved enzyme variants validated experimentally.

Frontier 2 — Elastane / Spandex: The Universal Contaminant

State of the art:

- Elastane (Lycra/Spandex/Creora/Roica/ESPA) is a polyether/polyester-polyurea segmented polyurethane. The hard segments resist mechanical shredding (cause filter blocking), block enzymatic depolymerization (block enzyme access to substrate), and generate volatile/crackling byproducts at melt-temperatures relevant for PET (240–280 °C — Mistra Future Fashion data). Adroitmarketresearch
- **Selective dissolution / degradation lab approaches:**
  - **DMSO + DBN** (1,5-diazabicyclo[4.3.0]non-5-ene catalyst): 81–100% elastane mass loss at 80–120 °C in 10–60 min (Phan/De Clerck/De Meester group, Ghent, 2023–2025). DMSO alone: 26–32% at 80–100 °C; 79% at 120 °C/10 min. This is the most advanced selective-dissolution work published. ResearchGate ResearchGate
  - **Chitosan dissolvable finish on elastane filaments** — applied at production, selectively removed at end-of-life; enables clean blend separation. Pre-commercial. ResearchGate
  - **Enzymatic PU degradation**: lacasse + mediator, cholesterol esterase, lipase systems. Chavez-Linares 2025 review: "However, the design of efficient enzymes still needs to be developed so that enzymatic depolymerization can be considered as a competitive method to mechanical and chemical ones." Reaction times still in 18+ days; enzymes typically work only on PU model substrates, not on technical elastane fibers. ResearchGate
  - **Glycolysis/aminolysis of PU foam**: scalable ionic-liquid-catalyzed glycolysis at 115 °C, 7 h with 95% recovered polyol (Green Chem 2025, 10.1039/D5GC03643G). Applied to PU foams, not yet to elastane fibers. RSC Publishing
- **Sciencewise, the open seam is selective deconstruction of urea-rich hard segments while preserving the surrounding polyester/polyamide/cotton fiber matrix**, ideally at room temperature, ideally with recoverable monomer (polyol + diisocyanate equivalent), ideally compatible with downstream PET/cotton processing.

Attack vector: Predominantly *chemistry*, but with one explicit software-substitute opportunity: an AI-driven generative model that can propose elastane-selective solvent systems, enzyme variants, or catalyst combinations against a defined cost/temperature/selectivity objective is genuinely open ground. The Samsara "AI platform" is the closest existing precedent.

Frontier 3 — Monomaterial / Design-for-Disassembly Textiles

State of the art:

- **Resortecs** (Brussels, 2017): 16 heat-dissolvable Smart Stitch™ threads (bio-based and synthetic) with melting points 150, 170, 190 °C. Smart Disassembly™ industrial-scale thermal-disassembly ovens. **AMANN (industrial thread manufacturer) partnership 2025** for mass-industrialization of the threads — this is the key commercial step. Recycling rates up to 90% on original fabric; 5× faster than manual disassembly per Circle Economy Foundation. Resortecs + 2
- **Mono-material synthetic stretch** is technologically real (mechanically textured PET → some stretch, no elastane) but the elasticity / recovery / fit performance vs. elastane is significantly worse — especially relevant for diapers, swaddles, leggings, growth-friendly waistbands in infant wear.
- **Reversible-chemistry / covalent-adaptable network** seams (Diels-Alder reversible thermosets, vitrimers for textile applications) are at TRL 2–4 (academic only).

Unsolved seam: A *thermoplastic-elastomer-as-a-mono-material* that delivers Lycra-equivalent stretch (300%+ elongation, 95% recovery) AND is depolymerizable AND survives 100+ wash cycles AND is dermally safe for infants. Carbon-negative bonus.

Attack vector: Chemistry (TPE design), with a real AI sub-opportunity in *polymer-property inverse design* — i.e., generative models for monomer/copolymer architectures targeting an elasticity-vs-recyclability Pareto front. This is a serious deep-learning project (training data is the chokepoint).

Frontier 4 — Microfiber / Microplastic Shedding (Source-Level)

State of the art:

- The literature (Khan et al. 2025; Carney Almroth lab; UToronto Golovin lab) is converging on three intervention layers: (a) yarn-level (finer, filament not spun, tightly twisted), (b) fabric-level (compact knit/weave, anti-pilling), (c) finishing-level (silicone, enzyme hydrolysis, PDMS coatings).
- Polysilk-CTE (silicone-based finish): 63.25% reduction on polyamide.
- UToronto's PDMS-shampoo coating (Golovin lab): explicitly aiming at multi-fiber-substrate adhesion — "We are working to formulate the correct polymer architecture so that our coating can durably adhere to all of those fibers simultaneously." WWD
- Enzyme hydrolysis pre-treatment on polyester knit: significant reduction over 20 wash cycles. PubMed Central
- France's mandatory washing-machine microfiber filter (Loi AGEC) effective Jan 1, 2025. Pollutec
- Under Armour: 75% low-shed fabrics by 2030; their internal Fiber Shed Test Method opened publicly Dec 2023. Textileweb

The measurement seam (the actual unsolved problem): No universally adopted ISO test; AATCC TM212 (under development), CEN/TC 248/WG 37, the Microfiber Consortium TMC test method, ASTM F08 working group, and AFNOR all exist but no single benchmark accepted by brands, regulators, NGOs. The As You Sow shareholder resolutions on Target (2026 filing, "26TGTCE") and Gap (Dec 2025) explicitly identify the lack of harmonized testing as the blocking issue for setting brand reduction targets.

The infant-skin specific concern: the *Int. J. Women's Health* 2025 systematic review (PMC12412761) and the 2025 *Sustainable Chemistry & Pharmacy* scoping review (S2214750025002628) both flag dermal contact through clothing as an underexamined route. AAP-cited 30% thinner skin in infants. Polyester is the most common indoor airborne microfiber polymer in 2024 studies.

Attack vector: Mixed. The coating-side is chemistry/materials. The measurement-side is software + sensing + standardization — this is the genuinely highest-shock opportunity (a benchmark dataset + standardized test that the entire industry coalesces around is field-defining, and no team has done it).

Frontier 5 — Textile Fiber Identification & Sorting Tech

State of the art:

- **Hyperspectral imaging (HSI):** Specim FX17 (900–1700 nm) is the industrial reference; PICVISA integrates it. ~224 wavelength channels per pixel. Specim + 2
- **Refiberd** (founded 2020, $3.4M seed via True Wealth Ventures): proprietary AI + HSI; claims trace-detection of spandex/nylon/acrylic <2% composition. The most aggressive published spec.
- **Matoha** (2017): handheld NIR scanners for manual sorting augmentation. Lower cost, lower accuracy. WIPO
- **Fibersort/Valvan, Tomra Autosort, Andritz/Pellenc, TrinamiX, Sortile, Picvisa** are the industrial-scale conveyor systems.
- **Siptex** (Malmö, SE): the only operating industrial-scale HSI sorting plant in Europe at scale. ~24,000 t/y design capacity.
- **2025 arXiv paper 2505.03575**: supervised and unsupervised deep learning (CNN) on NIR-HSI of textiles outperforms classical chemometrics. arxiv
- **2025 ScienceDirect S1386142525009722**: NIR-HSI explicitly flags low-elastane detection as challenging; reports limit of detection issues with carbon-black-dyed fabrics. ScienceDirect

Unsolved seams:

1. **Dark/black textile blindness** — carbon black absorbs across NIR. No accuracy benchmark above ~70% on black blends.

2. **Sub-2% elastane quantification** at field/MRF speed — Refiberd's claim is unaudited.
3. **No public benchmark dataset** for blended/contaminated/worn/dyed textile HSI — every commercial vendor uses proprietary internal data.
4. **No quantification of dyes/finishes/PFAS by HSI** — these are decision-critical for CPSIA-recycled feedstock but invisible in NIR.
5. **Raman + HSI fusion** is academically promising but not commercialized.

Attack vector: AI/sensing dominant. This is the area where a software-first team can plausibly produce a hyper-shocking demo: e.g., an open-source benchmark + foundation model for textile HSI; or a sub-NIR (LWIR/mid-IR/THz) sensing approach that defeats carbon black; or RGB-only blend prediction (transfer-learned from HSI labels) deployable on a phone. The chokepoint is access to labeled data.

Frontier 6 — Robotic / Automated Garment Disassembly

State of the art:

- **Soft-fabric manipulation** is an established hard problem in robotics. APS-Net (Liu et al., ICML 2025) uses dual-arm dynamic flinging + pick-and-place to achieve unfolding + standardization with a learned reward function. SoftGym/PyFleX is the simulation backbone. ICML
- **Hard-point removal** (zippers, snaps, buttons, embroidery): Carbios's textile preparation line (Nov 2023) integrates shredding + automated hard-point extraction in a single industrial line — but this is mechanical shred-and-screen, not surgical removal. Carbios
- **No published robotic system performs surgical disassembly** (cut along seam lines, remove zipper, separate panels) on a worn, deformable, dirty garment at industrial throughput. The closest academic work is the Shell-Type Soft Jig (arXiv 2509.13802, 2025) for small electronics disassembly.
- **Resortecs's heat-dissolvable threads** are the orthogonal route: if the thread is engineered to fail at 150–190 °C, no soft-manipulation is needed at all — the garment falls apart in a thermal disassembly oven. **This is a "chemistry sidesteps robotics" example.**

Unsolved seam: Real-world soft manipulation of garments under industrial throughput constraints; specifically, a robot that can identify a seam, follow it, cut it, and pull a zipper free in <30 seconds per garment.

Attack vector: Robotics + vision/AI. This is the *hardest* of the attack vectors to short-circuit with software alone, because the physical manipulation is the bottleneck. The strongest "software substitutes for chemistry" plays here would be (a) a foundation model for garment-state estimation under occlusion/deformation, or (b) a sim-to-real pipeline trained on synthetic infant garments with hard-points.

Frontier 7 — Chemical Decontamination of Recycled Feedstock

State of the art:

- **PFAS detection in polymer waste**: Whelan et al. UPLC-TripleTOF/MS method, 21 PFAS at 0.1–0.4 mg/kg LOQ (PMC9860823). Whitehead et al. *EST Letters* 2022: 100% of 72 stain-resistant US/Canadian children's textiles tested contained PFAS, with school uniforms among the highest measured concentrations. SofiaMila
- **PFAS destruction**: supercritical water oxidation (SCWO; Battelle, Aquagga, 374-Water), electrochemical oxidation, plasma destruction, hydrothermal alkaline treatment (HALT). None deployed on textile feedstock specifically.
- **AZO dye / azo cleavage**: aerobic-anaerobic bioreactor systems can cleave azo bonds to aromatic amines (which are themselves regulated). No commercial system targets recycled textile feedstock.
- **Lead, phthalates, flame retardants in legacy children's textiles**: the EU POP regulation imposes Low POP Concentration Limit values; textile recyclers must screen, but **no in-line continuous-stream screening technology exists**.
- **CPSIA gap:** CPSIA mandates lead, phthalate, ASTM F963 testing for children's products; does not currently mandate PFAS or PBDEs. State laws (CA, NY, ME, OR TFKA) layer on. **The unsolved gap: there is no economic, in-line, MRF-speed sensor for any of these chemicals.** They are all currently caught at the sample-and-lab step.

Unsolved seam: A real-time, in-line PFAS/phthalate/dye/heavy-metal detection sensor at MRF speed. This is the analog of NIR for fibers but for legacy chemicals — and it does not exist.

Attack vector: Sensing + AI. Spectroscopy (LIBS for metals, FT-IR or surface-enhanced Raman for organics, mass-spec-on-a-chip) + AI for classification. Genuinely open ground. The wet-chem alternative (rinse-and-extract sampling at MRF stages) is technically solved but unscalable.

Frontier 8 — Advanced / Chemical Recycling of Plastics (Cox/Nexus Thesis)

State of the art:

- **Nexus Circular** (Atlanta): pyrolysis of HDPE/LDPE/PP/PS films at the 33 M lb/y Atlanta facility; ~80% oil yield on input; ISCC PLUS certified since 2018. Backed by Cox Enterprises (lead investor, majority owner after $150M 2024 raise), Braskem, Chevron Phillips Chemical, Shell, Printpack. Pipeline: Dallas (26,000 t/y), Chicago (30,000 t/y expandable to 120,000 t/y, Braskem offtake). 600,000 sq ft plant in construction. Recycling Today Packaging Dive
- **Mura Technology HydroPRS™** (supercritical water, not pyrolysis): 20,000 t/y Teesside (ReNew ELP) operational late 2023/2024. 30-min reactor residence. Offtake: Dow, Neste, LG Chem, CPChem. JRC LCA: ~50% lower GWP vs pyrolysis. 1.5 Mt/y target 2032 via KBR licensing. Igus + 4

- **Plastic Energy TACOIL**: 85% reported yield. Almeria + Sevilla operational since 2017. SABIC JV Geleen 20,000 t/y commissioning end-2024. TotalEnergies Grandpuits 15,000 t/y. ExxonMobil Port Jérôme 25,000 t/y scaling to 33,000 t/y. Resourcewise + 3
- **ExxonMobil Baytown** (proprietary process + Cyclyx feedstock): cumulative >100 M lb processed by May 2025. Target 500 M lb/y US by end-2026 (Baytown + Beaumont, six units); 1 B lb/y globally by 2027. $200M+ investment announced Nov 2024. ExxonMobil + 3
- **BASF ChemCycling**: mass-balance ISCC+. Partners Quantafuel, Pyrum (€16M invested), New Energy, Braven Environmental (March 2025 offtake for Port Arthur). BASF + 2
- **Eastman methanolysis** (Kingsport, TN): opened March 2024 — world's largest molecular recycling. PET focus. 2025: 105% nameplate; 2.5× the 2024 volumes; ~$60M incremental earnings. Longview TX second plant delayed 2 years after May 2025 DOE $375M grant cancellation. U.S. Plastics Pact + 3
- **Brightmark** (Ashley, IN): the cautionary case. 100,000 t/y nameplate. Operating at five percent capacity per March 2025 bankruptcy filings. Chapter 11 March 17, 2025 ($178.35M secured debt, $172.5M green bonds). Sold back to parent at $14.3M vs $260M construction cost. Thomaston, GA $950M / 400,000 t/y project announced May 2024 reportedly unaffected. Resource Recycling + 8

The three contested seams:

1. **End-to-end yield.** Per Lisa Song, *ProPublica* (Jun 2024): "if a pyrolysis operator started with 100 pounds of plastic waste, it can expect to end up with 15–20 pounds of reusable plastic," and 25–30 pounds for advanced technologies. ZWE *Leaky Loop* Oct 2023: "at most 2%." Industry oil yields (Nexus ~80%, Plastic Energy 85%) are mass-to-oil, not mass-to-new-plastic; the steam-cracker step recovers <50% of oil to ethylene/propylene. Sustainable Plastics
2. **The chlorine/PVC/heteroatom ceiling.** Strien et al. *Energy & Fuels* 2024 (10.1021/acs.energyfuels.4c04745): 78% oil from dry-washed DKR-350 feedstock with measurable Cl/N/S residuals. Lee et al. *Phil. Trans. R. Soc. A* 2025: "none has been proven at the commercial scale" for PVC dechlorination. HCl is autocatalytic and poisons Pt/ZSM-5 downstream catalysts (*Sci. China Chem.* 2024). Multi-layer films with EVOH, nylon barrier, and aluminum-deposited PET introduce N, O, metal contaminants. nih Springer
3. **Mass-balance accounting ("free allocation" vs "fuel-use exempt" vs "polymer-only" vs "proportional").** NRDC (Renée Sharp): "you can use feedstock that is only 10 percent plastic waste and 90 percent virgin material and market the products as having 100 percent recycled content." Beyond Plastics/IPEN *Chemical Recycling: A Dangerous Deception* (Oct 2023, Lee Bell): "Only 11 chemical recycling facilities currently exist in the United States, and in total they are capable of processing less than 1.3 percent of all plastic waste generated annually." Jan Dell (Last Beach Cleanup): if a brand claims >2% recycled content via ISCC Plus, "somewhere in the brand's approved accounting system plastics are being turned into fuels." California AG sued ExxonMobil in 2024 explicitly criticizing ISCC mass-

balance. The Cefic/20-association joint letter (2024) lobbies for "mass balance fuel use exempt" as harmonized EU method: "An estimated 8-billion-euro investment is in the pipeline by 2030 that would produce 2.8Mt recycled plastics via chemical recycling… Other methods such as polymer only and proportional allocation will significantly increase the costs." NRDC + 5

The catalytic frontier (where software/AI matters most):

- **Sodium-on-alumina + WO_3/SiO_2** (Conk/Hartwig, *Science* 2024, 10.1126/science.adq7316): PE/PP → propylene + isobutylene at ~90% lab yield. Susannah Scott commenting in C&EN: "This is a pretty big step forward." ScienceDaily ACS C&EN
- **$Pt/SrTiO_3$** (Celik/Sadow/Delferro, *ACS Cent. Sci.* 2019): PE → narrow MW liquid products at 300 °C, 170 psi H_2. Bepress
- **Mesoporous-silica-encapsulated Pt** (Tennakoon/Sadow/Perras, *Nat. Catal.* 2020): nanopore "molecular ruler" architecture controlling chain length. C&EN
- **PE → propylene tandem ethenolysis-isomerization** (Wang/Strong/Scott/Guironnet, *JACS* 2022, 144:18526-18531): one-pot PE-to-monomer. ChemRxiv
- **PE → alkylaromatics** (Zhang/Scott/Abu-Omar, *Science* 2020, 370:437-441): one-pot PE-to-aromatic-detergent precursors. Ameslab
- **2026 ML breakthrough** (Byron/Sadow/Delferro/Ferrandon et al., *ACS Catalysis* 2026, 16(1):431-445): "Data-Driven Discovery of Bimetallic Nanoparticles Catalysts for the Hydrogenolysis of Polyethylene." This is the first explicit ML-discovered plastic-deconstruction catalyst. ResearchGate ACS Publications
- **Mechano-catalysis** (Hergesell/Weckhuysen/Vollmer, *JACS* 2024, 146(38):26139-26147): ambient-temp ball-mill deconstruction.
- **Ru-quasi-supercritical hydrogenolysis** (Kim/Lercher, *JACS Au* 2025, 10.1021/jacsau.5c00006): industrial-academic parallel to Mura. nih

Textile-pyrolysis intersection: Direct pyrolysis of polyester textiles is essentially zero at commercial scale. The blockers: dyes generate phthalate/disperse-dye-derived sulfur and nitrogen heteroatoms in oil; elastane generates polyurethane-derived nitrogen contaminants; PVC prints generate chlorine; PU coatings generate nitrogen. The commercial textile pathway is depolymerization (Carbios, Eastman, Syre, FENC) not pyrolysis. **There is an open opportunity to pre-process textile waste into a pyrolysis-grade feedstock by selectively removing dyes/elastane/PU coatings upstream — but no commercial system does this today.** GlobeNewswire Research And Markets

Attack vector: Mass-balance/accounting is *software* (a credible third-party verification platform is genuinely high-shock). The catalyst frontier is *AI-driven inverse design + wet-lab validation* — the Byron/Sadow 2026 paper is the existence proof.

Frontier 9 — Biomanufactured / Next-Gen Materials for Infant-Safe Apparel

State of the art:

- **Spiber Brewed Protein** (Tsuruoka, JP): precision-fermented protein fibers. Commercial-scale (Thailand, ~400 t/y). North Face Moon Parka, Pangaia, Goldwin collaborations. TRL 8.
- **Keel Labs Kelsun** (NC): kelp-derived alginate fiber. TRL 6–7. Outerknown, Stella McCartney prototypes.
- **Newlight AirCarbon** (CA): methane-fed PHA biopolymer. TRL 7. Apparel pilots.
- **MycoWorks Reishi, Bolt Threads Mylo** (now discontinued), Ecovative Forager: mycelium-based leather alternatives. Reishi commercial w/ Hermès, Ligne Roset. Mostly leather replacement, not knit apparel.
- **Modern Synthesis**: bacterial nanocellulose textiles. Pre-commercial.
- **Colorifix** (Norwich, UK): microbial precision-fermented dyes. $18M Series B June 2025 led by Inter IKEA Group (per *Just-Style*, Jun 24, 2025), with H&M Group Ventures as continuing investor. **First biological dye with OEKO-TEX Eco Passport that includes biocompatibility tests for skin allergenicity, skin irritation, and cytotoxicity** — the only published biocompatibility-tested next-gen colorant. TRL 7. Just Style Colorifix
- **Pili Bio** (FR): bio-based aniline/indigo. TRL 5–6.
- **Living Ink Algae Black, Rubi Laboratories CO_2-to-cellulose, Newlight, Pangaia, On collaborations.**

The CPSIA/REACH/infant-safety gap: CPSIA mandates lead (⩽100 ppm in substrate), phthalates (DEHP/DBP/BBP/DINP/DIDP/DnOP/DIBP/DPENP/DHEXP/DCHP at ⩽0.1%), and ASTM F963 mechanical/flammability. The TFKA (Oregon), CA Prop 65, Maine PFAS-in-products, NY child-product chemicals law layer on. **Next-gen biomaterials are not in the published CPSIA-tested cohort for infant apparel; brands using them in baby/toddler categories carry independent toxicology liability.** A team that publishes a CPSIA-and-REACH-compliance toxicology framework for next-gen materials in infant apparel would create a new public good.

Attack vector: Chemistry/biotech for the materials themselves; software for regulatory/toxicology framework synthesis.

Frontier 10 — Demand-Side / Overproduction / Rebound

State of the art:

- **The Yerushalmi 2025 finding** (*Business Strategy and the Environment* 10.1002/bse.70135): "Our findings reveal a 155% rebound backfire, showing that CE innovations in the TC sector may exacerbate rather than mitigate environmental pressures." This is a multi-region, multi-sector dynamic computable general equilibrium (DCGE) model — the most rigorous quantitative CE rebound finding to date for textiles. Wiley Online Library

- **Zink & Geyer 2017** (*J. Industrial Ecology*) framed the Circular Economy Rebound concept; ~700 citations.
- AI demand-forecasting (Inditex, H&M, Shein internal systems) has *not* visibly reduced global apparel overproduction; instead, the speed/ultra-fast cycle (Shein adds roughly 2,000 new SKUs/day per Business of Apps 2026, with Sacra 2025 putting it in the 1,000–3,000 range) has been accelerated.
- **Real-time manufacturing-capacity-reservation systems** (an emerging concept where brands reserve mill capacity rather than ordering by SKU) — academically discussed (MIT Sloan, BCG) but not commercially implemented.
- **Pre-order / made-to-order at scale** (Unspun, On Pre-order pilots): TRL 6–7 in adult apparel, almost zero in children's apparel because growth-driven sizing complicates fit.

Unsolved seam: A demand-side mechanism that demonstrably reduces total apparel volume produced, validated with a transparent model rather than brand self-report. The DCGE methodology used by Yerushalmi could be open-sourced and applied to a specific brand's portfolio.

Attack vector: Pure software/systems-modeling. This is the highest-ceiling software-only frontier — and the one most easily dismissed as "not a hackathon project." A team that builds a credible rebound-effect simulator for a real apparel portfolio is doing serious sustainability science.

Frontier 11 — Physical-Layer Traceability

State of the art:

- **Haelixa** (ETH Zürich spinout 2016): encapsulated plant-DNA markers, PCR-detectable. Survives spinning, weaving, dyeing, bleaching, finishing, and unlimited wash cycles. Does NOT survive gold/silver melting or surface-layer removal. Production deployments: New Focus Textiles (Nov 2025, recycled cotton denim), Soorty SecondLife (2024, recycled cotton denim), C&A, Hugo Boss, OVS. OEKO-TEX & GOTS compliant. Global Venturing + 4
- **FibreTrace** (woven luminescent pigments): scannable at every supply-chain stage. Reformation, 7 For All Mankind. WWD
- **Oritain** (forensic origin verification via isotope fingerprinting): Cone Denim, Primark. Does not survive recycling. WWD
- **TextileGenesis, TrustTrace** (digital-twin/blockchain): software-layer only. Haelixa
- **GenuTrace** (isotope fingerprinting for fiber-origin verification): TRL 6.
- **Woven RFID** (TexTrace, Smartrac): durable through wash, but bulky and not infant-skin-friendly.

Unsolved seams:

1. **A marker that survives both 100+ washes AND chemical recycling.** No commercial system does this.
2. **A marker readable by consumer-grade hardware** (smartphone, not lab PCR).
3. **A marker that survives shredding for mechanical recycling and is recoverable in the regenerated fiber** — so that recycled-content claims can be physically verified at the new product.
4. **A marker that can encode end-of-life processing instructions** (best route: enzymatic/mechanical/incinerate).

Attack vector: Materials chemistry for the marker itself; **AI/computer vision** for reading. The frontier intersection: spectral/fluorescent taggants designed for sub-second readout at MRF speeds, combined with foundation-model-class spectral inference. This is genuinely open.

Frontier 12 — Other Galaxy-Tier Open Problems

- **Closed-loop waterless dyeing & dye recovery:** ColorZen (cotton pre-treatment, ~95% water reduction), DyeCoo CO_2-dyeing (Adidas DryDye), Imogo Dye-Max (95% water reduction). The frontier gap: dye-recovery from rinse water at gigaton scale; dye-as-monomer (dyes that double as polymer building blocks).
- **CO_2-derived fibers:** LanzaTech CarbonSmart polyester in commercial Lululemon/On/Zara/H&M Move products since 2024. Twelve+LanzaTech CO_2-to-polypropylene 2024. The frontier gap: direct CO_2-to-fiber (skip the gas-fermentation step). Solidia, CarbonCure-style carbonate chemistry on textile-grade polymers is wholly open ground. Lanzatech
- **Fiber-to-fiber yield degradation over multiple cycles:** Chemical recyclers (Carbios, Eastman, Samsara) claim virgin-equivalent; mechanical recyclers (RE&UP, New Focus) acknowledge ~30% degradation per cycle. **The thermodynamic limit of cellulose recycling** (DP, degree of polymerization, falls every cycle) is the underlying physics problem; no commercial process has solved it.
- **Economic/thermodynamic limits of recycling:** Recycled feedstock typically trades at a roughly 30% premium over virgin per S&P Global Commodity Insights (Jul 2025), with peaks of ~1.7× during tight markets (post-consumer natural HDPE reached ~$1,631/tonne vs. virgin HDPE at $943/tonne in summer 2023, per Platts data reported by Resource Recycling, Aug 2023). **Hartz (Nexus): "If we wanted to pay $1 a pound, we can get all the feedstock we want. But we will blow up our economics. We average about 5 to 8 cents per pound."** This is the economic seam: recyclers cannot pay enough for feedstock to drive collection, so collection rates stay low.

Recommendations

For an ambitious hackathon team chasing maximum technical ceiling, the highest-shock-per-effort zones, ranked:

Tier 1 — Bleeding-edge software-attacks-chemistry intersections (highest upside, hardest to execute well):

1. **AI-driven inverse design of elastane-selective solvents or catalysts.** Take the Byron/Sadow 2026 ML-bimetallic-catalyst-discovery framework and re-target it to PU/elastane segments. Use AlphaFold/RoseTTAFold or molecular generative models (MOFs/COFs/zeolites) constrained to a polyether-polyurea cleavage objective. The Samsara enzymatic-design pipeline is the closest commercial precedent. **Threshold to change tier:** if a team can demonstrate a wet-lab validated novel solvent system, that's a *Nature*-scale result.
2. **Open foundation model + open benchmark dataset for textile HSI with elastane sub-2% quantification on dark/dyed fabrics.** The data scarcity is the chokepoint; partnership with Refiberd/Matoha/Siptex/Carbios for labels would gate this. **Threshold:** if a team can demonstrate >85% F1 on a publicly held-out test set with dark blends, this is field-defining.
3. **Real-time MRF-speed spectral sensor for legacy chemicals** (PFAS, phthalates, AZO dyes, Pb) in textile recycling streams, AI-classified. No commercial system exists. **Threshold:** detection of named PFAS at 0.1 mg/kg in moving feedstock.

Tier 2 — Systems-level audacity (highest "absurdly complete platform" wow-factor):

4. **Open-source rebound-effect simulator** for an apparel portfolio (DCGE-based, following Yerushalmi 2025 methodology) that quantifies for a specific brand the rebound risk of a specific circular intervention. This is the only frontier where a software team alone can produce honest, publishable science about why fashion sustainability is failing.
5. **End-to-end physical-marker-to-chemistry-survival traceability system:** a marker chemistry that survives 100+ washes AND enzymatic/glycolysis depolymerization AND is recoverable in the regenerated fiber. The Haelixa-survives-wash + a depolymerization-stable secondary marker (perhaps an isotope-labeled monomer fraction, e.g., ^{13}C-enriched TPA introduced at <0.1%, detected by IRMS in recycled output) is the proposed unsolved architecture.

Tier 3 — Highest wet-lab dependency, hardest to do in 72 hours, but the "this shouldn't be possible" demo zone:

6. **A working bench-scale elastane-selective dissolution + recovery rig** following the Phan/De Meester DMSO/DBN protocol, demonstrating recovery of polyol monomer as a usable input.

7. **A microfiber-shedding test standard + reference fabric set** — a public benchmark (open garments + open data + open protocol) that the industry could adopt. This is the field-defining gap.

Benchmarks that would change the recommendations:

- If a team has wet-lab access and a synthetic chemist: prioritize Tier 1.1 or Tier 3.6.
- If a team is software-only with no lab: prioritize Tier 1.2 or Tier 2.4.
- If sensing hardware is available: Tier 1.3.
- If the team's instinct is "absurd platform completeness": Tier 2.5 is the highest ceiling.

Caveats

1. **Industry yield claims are systematically optimistic.** Pyrolysis oil yields (80–85%) are mass-to-oil, not mass-to-new-plastic. ProPublica's 15–30% end-to-end yield figure is independently synthesized; Zero Waste Europe's 2% figure is the most aggressive critic number but draws on 22 peer-reviewed papers. Both Nexus Circular and Plastic Energy are real industrial operators; Brightmark's bankruptcy is the cautionary counterexample.
2. **Mass-balance accounting is genuinely contested** and a hackathon team that confidently builds anything on top of ISCC PLUS recycled-content claims should explicitly disclose the methodology and the unresolved fuel-use-exemption controversy. The California AG v. ExxonMobil litigation is ongoing.
3. **The Refiberd <2% elastane detection claim is company self-report**; no independent peer-reviewed validation exists. Multiple academic NIR-HSI papers in 2025 (ScienceDirect S1386142525009722) explicitly note that low-elastane detection remains challenging.
4. **Several "next-gen materials" claims (Spiber, Keel Labs, Newlight, Modern Synthesis) are at low commercial volume and have not been independently CPSIA-tested for infant apparel.** Brands using them in baby/toddler categories carry independent toxicology obligations.
5. **The rebound-effect 155% finding** (Yerushalmi 2025) is one DCGE study with specific assumptions; sensitivity analysis would be needed before treating it as a settled empirical finding. The qualitative direction (rebound is real in textiles) is well-established; the precise magnitude is one paper.
6. **No commercial enzyme has been demonstrated to degrade elastane in a real garment matrix at industrial conditions.** The published TRL is generously 3–4 for elastane-targeted enzymology. The DMSO/DBN selective-dissolution chemistry is closer to commercial than enzymatic degradation as of mid-2026.
7. **Haelixa DNA markers will not survive enzymatic depolymerization or pyrolysis.** Brand claims that "recycled content is verifiable" need to distinguish "verifiable in the cotton-yarn

supply chain" from "verifiable in the chemically recycled monomer stream" — these are different verification problems.

8. **CPSIA does not currently require PFAS testing.** State-level requirements vary (CA, NY, ME, OR, WA) with effective dates spreading 2024–2027. A hackathon platform built on "PFAS-free recycled feedstock" must reference state law specifically, not CPSIA broadly.